

Buoyancy-Stratified Factorial Geometry — Riemann Curvature of the Aether Metric

Daniel T. Murphy

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PAPER_554: Buoyancy-Stratified Factorial Ge- ometry — Riemann Curvature of the Aether Metric

Author: Daniel T. Murphy — Star Magic / UQFF Framework

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CP4 Class: BSFGriemannCurvatureAetherMetricCalculator (#149)

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Context note: The Aether metric $A_{\mu\nu}$ was introduced in PAPER_392 (CP4 #43) with coupling $\eta = 10^{-22}$ and scalar amplitude T_{s00} . PAPER_416 (CP4 #66) provided the five-component spatial decomposition of $T_{s00}(r)$. This paper combines those results to derive for the first time the full Riemann curvature tensor of the BSFG geometry — the first paper in the BSFG Complete Geometric System series (PAPER_554–558).

Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Riemann Curvature of the Aether Metric, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

The Buoyancy-Stratified Factorial Geometry (BSFG) is defined on a 26-dimensional pseudo-Riemannian manifold $\mathcal{M}^{26}$ equipped with the Aether-perturbed metric $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r) \delta_{\mu\nu}$, where $\varepsilon(r) = \eta, T_{s00}(r) \cos(\pi t_n)$ and $T_{s00}(r)$ is the five-component stellar stress-energy density from PAPER_416.

This paper derives the Christoffel symbols, Riemann curvature tensor, Ricci tensor, and Ricci scalar for the 4D slice of BSFG at a field point r . The key result is:

$$R^r_{\ 0r0} \approx \frac{\varepsilon''}{2} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

with $C_{\text{num}} = (M_s c^2 + L_s/c^2)/(4\pi/3)$. At the solar surface $r = R_\odot$: $R^r_{\ 0r0} \approx 1.56 \times 10^{-19} \text{ m}^{-2}$, which is $\approx 4 \times 10^{-26}$ times the Schwarzschild curvature — confirming the Aether contribution is a genuine but ultra-weak geometric perturbation.

§2 The Aether Metric — Previous Results

From PAPER_392 (CP4 #43), the BSFG metric is:

$$A_{\mu\nu} = g_{\mu\nu} + \eta, T_{s00} \cos(\pi t_n) \delta_{\mu\nu}$$

In components (4D Minkowski background with signature $+---$):

$$A_{\mu\nu} = \text{diag}(1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon)$$

where $\varepsilon = \eta, T_{s00}(r) \cos(\pi t_n)$. From PAPER_416 (CP4 #66), the five-component Ts00 with dominant radial term:

$$T_{s00}(r) = \underbrace{\frac{M_s c^2}{\frac{4}{3}\pi r^3}}_{T_{-1} \propto r^{-3}} + \underbrace{\frac{L_s}{c^2 \cdot \frac{4}{3}\pi r^3}}_{T_{-2} \propto r^{-3}} + \underbrace{\frac{\rho_{SCm} v_{SCm}^2}{c^2} + \frac{\rho_A v_{UA}^2}{c^2} + \rho_{sw} v_{sw}^2}_{\text{constant terms}}$$

The radially-dependent numerator is $C_{\text{num}} = (M_s c^2 + L_s/c^2)/(4\pi/3) \approx 4.27 \times 10^{46} \text{ J} \cdot \text{m}$.

§3 Christoffel Symbols

Step 1. Compute $\varepsilon(r)$ and its radial derivatives:

$$\varepsilon'(r) = \frac{d\varepsilon}{dr} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{r^4}, \quad \varepsilon''(r) = +\frac{12\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

Step 2. For a diagonal metric $A_{\mu\mu}(r)$ depending only on $r = x^1$, the non-zero Christoffel symbols of the Levi-Civita connection are:

$$\Gamma_{\mu\mu}^r = -\frac{\partial_r A_{\mu\mu}}{2 A_{rr}} = -\frac{\varepsilon'}{2(-1 + \varepsilon)} \quad (\text{no sum on } \mu)$$

$$\Gamma_{\alpha r}^\alpha = \Gamma_{r\alpha}^\alpha = \frac{\partial_r A_{\alpha\alpha}}{2 A_{\alpha\alpha}} \quad (\text{no sum on } \alpha)$$

Explicitly at leading order in ε :

Symbol	Exact	Leading order
Γ_{00}^r	$-\varepsilon'/(2(-1 + \varepsilon))$	$+\varepsilon'/2$
Γ_{rr}^r	$+\varepsilon'/(2(-1 + \varepsilon))$	$-\varepsilon'/2$
Γ_{0r}^0	$+\varepsilon'/(2(1 + \varepsilon))$	$+\varepsilon'/2$
Γ_{ir}^i	$+\varepsilon'/(2(-1 + \varepsilon))$	$-\varepsilon'/2$
Γ_{ii}^r	$-\varepsilon'/(2(-1 + \varepsilon))$	$+\varepsilon'/2$

§4 Riemann Curvature Tensor

Step 3. Apply the Riemann tensor formula:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu u \Gamma^\rho_{\nu\sigma} - \partial_\nu u \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

The dominant component (tidal force in the radial-temporal plane):

$$R^r_{0r0} = \partial_r \text{Gamma}^r_{00} + \Gamma^r_{rr} \Gamma^r_{00} - \Gamma^r_{00} \Gamma^0_{0r}$$

$$= \frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{2} + O(\varepsilon^3) \approx \frac{\varepsilon''}{2}$$

Step 4. Substituting $\varepsilon'' = 12\eta \cos(\pi t_n) C_{\text{num}}/r^5$:

$$R^r_{0r0} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

The higher-order $(\varepsilon')^2$ correction is of order $\eta^2 T_{s00}^2 \sim 10^{-30}$, negligible.

§5 Ricci Tensor and Ricci Scalar

Ricci tensor — applying $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$ with $SO(3)$ spherical symmetry ($x^2 = y$, $x^3 = z$ contribute equally to $x^1 = r$):

$$R_{00} = R^r_{0r0} + R^\theta_{0\theta0} + R^\phi_{0\phi0} = 3 R^r_{0r0}$$

$$R_{rr} = -R^r_{0r0} + 2\left(\frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{4}\right)$$

Ricci scalar:

$$R = A^{\mu\nu} R_{\mu\nu} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} + \frac{2R_{\theta\theta}}{A_{\theta\theta}}$$

Kretschner scalar (leading order):

$$K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \approx 12 (R^r_{0r0})^2$$

§6 Numerical Values at the Solar Surface

At $r = R_\odot = 6.96 \times 10^8$ m, $t_n = 0$ (maximum coupling), $\eta = 10^{-22}$:

Quantity	Value	Notes
$\varepsilon(R_\odot)$	1.27×10^{-2}	Not small — linearization is qualitative at surface
$\varepsilon'(R_\odot)$	$-5.47 \times 10^{-11} \text{ m}^{-1}$	Aether gradient
$\varepsilon''(R_\odot)$	$+3.13 \times 10^{-19} \text{ m}^{-2}$	Curvature driver
R^r_{0r0}	$+1.56 \times 10^{-19} \text{ m}^{-2}$	BSFG tidal curvature
R_{scalar}	$\approx +3.0 \times 10^{-19} \text{ m}^{-2}$	de Sitter (gravity-dominant)
K	$\approx 2.9 \times 10^{-37} \text{ m}^{-4}$	Kretschner scalar

For comparison, the Schwarzschild curvature at $R_\odot$: $R^r_{0r0}|_{\text{GR}} \approx GM_\odot/R_\odot^3 \approx 3.95 \times 10^{-7} \text{ m}^{-2}$, giving:

$$\frac{R^r_{0r0}|_{\text{BSFG}}}{R^r_{0r0}|_{\text{GR}}} \approx \frac{1.56 \times 10^{-19}}{3.95 \times 10^{-7}} \approx 3.9 \times 10^{-13}$$

The BSFG Aether curvature is $\sim 10^{12}$ times weaker than GR at the solar surface, consistent with the perturbative assumption $\eta \sim 10^{-22}$ being a tiny coupling.



Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for $|\text{[SSq]}| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **curvature-D5** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{curv}})(\partial^\mu \phi_{\text{curv}}) - V(\phi_{\text{curv}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{curv}}) = \frac{1}{2}m^2\phi_{\text{curv}}^2 + \frac{\lambda}{4!}\phi_{\text{curv}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{curv}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{curv}}} = k_{\text{curv}} r_c^2 \cdot \partial_{D_5}(D_1 D_2 D_3 D_4 \cdot D_5) = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{curv}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.184 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 47, \quad n_{\text{channel}} = 9/26$$

Since $p_{\text{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **Hubble time** (super-Hubble saturation):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.184	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 47$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow g_{tt} = -(1 - 2\mu_s \nabla(M_s/r) \cdot r / c^2) \equiv \text{GR in } \varepsilon_{\text{BSFG}} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	PASS BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow \Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times (1 + \varepsilon_{\text{correction}})$	Cassini: $\Delta t / \Delta t_{\text{GR}} = 1 \pm 2.3\text{e-}5$	Cassini/GPP 2003	PASS Within Shapiro bound
Gravitational wave speed v_{GW}	BSFG: $v_{\text{GW}} = c \times (1 + k_{\eta}^2) \approx c + 10^{-226} \text{ m/s}$	GW150914 / GW170817: $ v_{\text{GW}}/c - 1 < 10^{-15}$	LIGO/Fermi GBM	PASS UQFF deviation 10^{-211} orders below bound
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\phi = \kappa \times \phi_{\text{GR}} \sim 10^{-6} \text{ arcsec/century}$	GR prediction: $43.03''/\text{century}$; observed: $43.1''$	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10^{-6} \text{ arcsec/century}$ to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§7 Connection to UQFF Framework

The Riemann tensor R^r_{0r0} of the BSFG geometry provides the **geometric encoding of tidal UQFF forces**. The buoyancy force F_U^{bi} represents the differential curvature between interior ($r < R_*$) and exterior ($r > R_*$) regions — a curvature discontinuity at the stellar boundary. The Aether field $\varepsilon(r) \propto r^{-3}$ creates a genuine non-flat geometry whose curvature $\propto r^{-5}$ decays rapidly away

from the source, consistent with the UQFF fifth-force measurements reported in PAPER_413–418.

Hub: PAPER_554 (#149) is the first paper in the BSFG series. See PAPER_558 (#153) for the complete geometric system definition and PAPER_555 (#150) for metric compatibility and geodesic equation.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	$F_{\{U_{Bi_i}\}}$ Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

5 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	Hybrid neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] * n / 26)$
<code>kozima_{wstp_kernel}.py</code>	Symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_26.py}</code>	426/26/[SSq] via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{fstp\kernel}.py</code>	Symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer (a;q)_n

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_hybrid}.py</code>	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}]^{\exp(-kappa_i*n/26)}]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_{1\CoAnQi}.cpp*, and Wolfram kernels (*uqff_{kozima_kernel}.wl*, *uqff_{s26_kernel}.wl*, *uqff_{mock_theta_pi_kernel}.wl*).

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